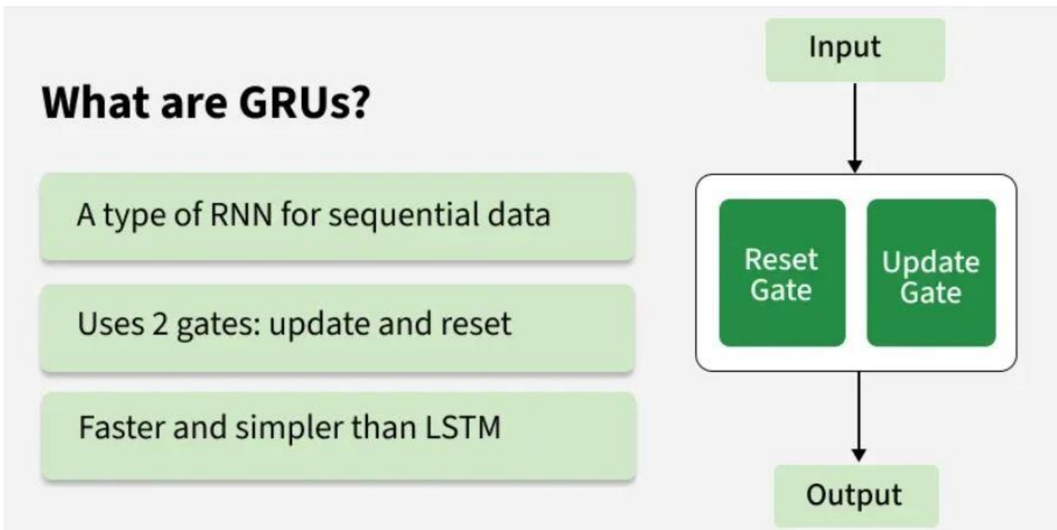
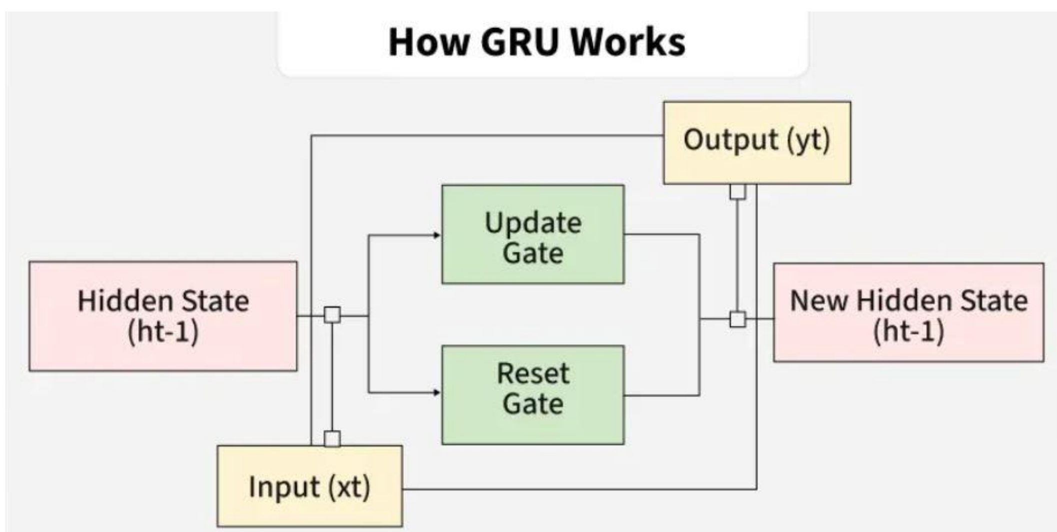
Gated Recurrent Unit Networks

Gated Recurrent Unit (GRU) networks are a type of recurrent neural network designed to handle sequential data while reducing the complexity of traditional RNNs. GRUs are a simplified advancement of LSTM, where they merge multiple gates into update and reset gates, hence learning long-term dependencies with faster training and fewer parameters.

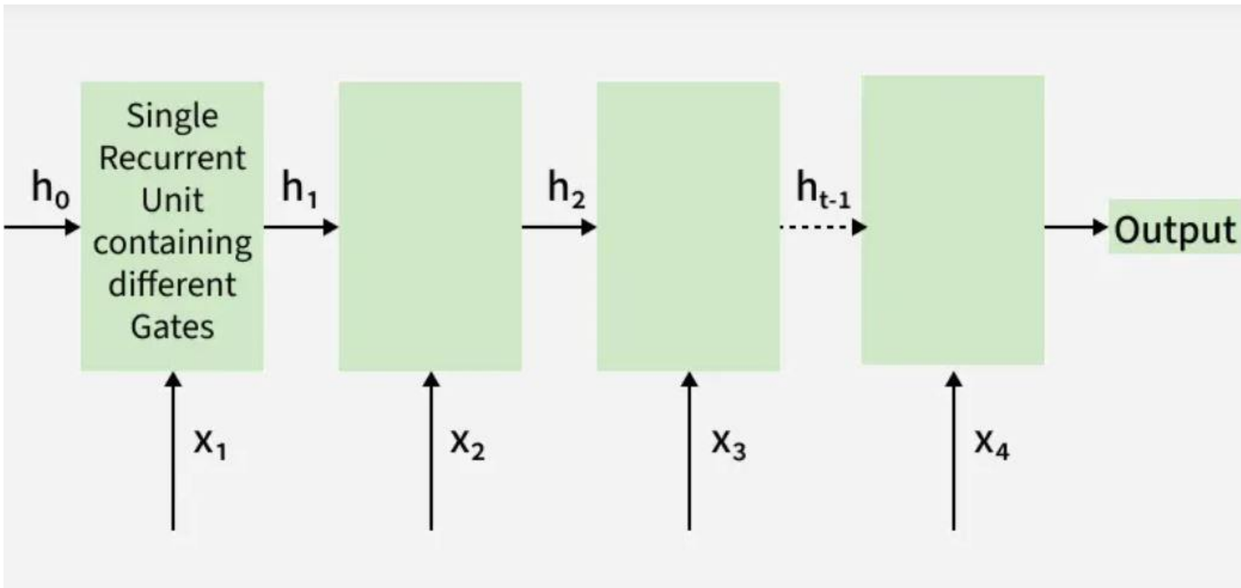
- Simplified alternative to LSTM
- Handles sequence and time-series data effectively
- Widely used in NLP, speech and forecasting tasks





What are Gated Recurrent Units (GRU)?

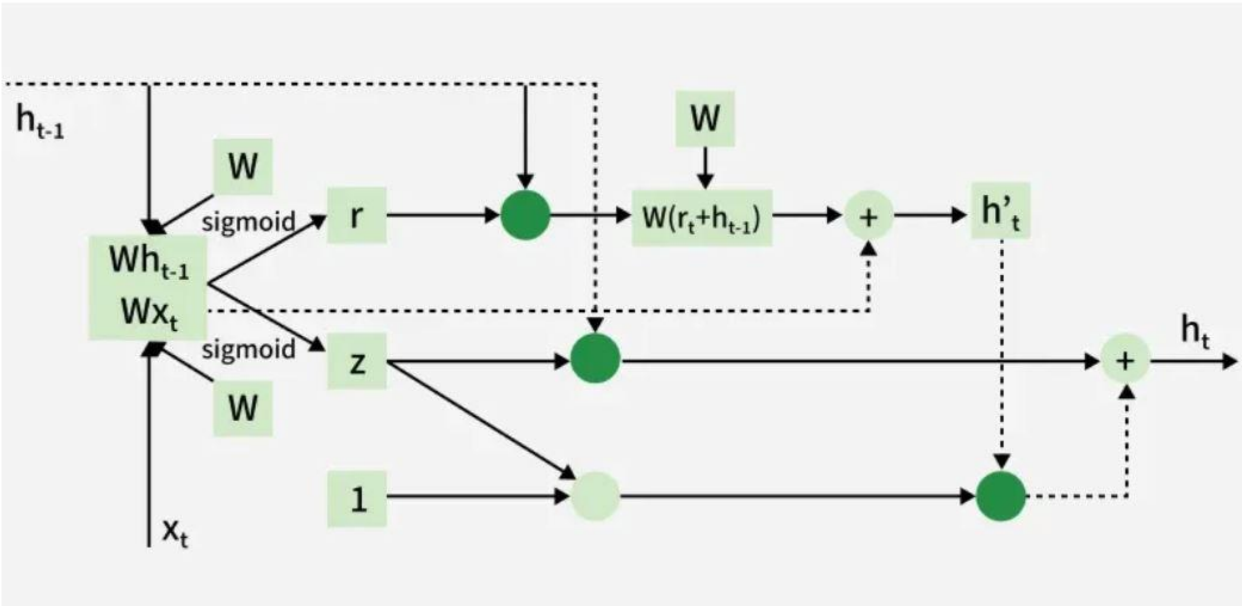
Gated Recurrent Units (GRUs) are a type of RNN introduced by Cho et al. in 2014. The core idea behind GRUs is to use gating mechanisms to selectively update the hidden state at each time step allowing them to remember important information while discarding irrelevant details. GRUs aim to simplify the LSTM architecture by merging some of its components and focusing on just two main gates: the update gate and the reset gate.



The GRU consists of two main gates:

1. **Update Gate** (zt*zt*): This gate decides how much information from previous hidden state should be retained for the next time step.
2. **Reset Gate** (rt*rt*): This gate determines how much of the past hidden state should be forgotten.

Architecture of GRUs



GRU vs LSTM

GRUs are more computationally efficient because they combine the forget and input gates into a single update gate. GRUs do not maintain an internal cell state as LSTMs do, instead they store information directly in the hidden state making them simpler and faster.

Feature	LSTM (Long Short-Term Memory)	GRU (Gated Recurrent Unit)
Gates	3 (Input, Forget, Output)	2 (Update, Reset)
Cell State	Yes it has cell state	No (Hidden state only)
Training Speed	Slower due to complexity	Faster due to simpler architecture

Feature	LSTM (Long Short-Term Memory)	GRU (Gated Recurrent Unit)
Computational Load	Higher due to more gates and parameters	Lower due to fewer gates and parameters
Performance	Often better in tasks requiring long-term memory	Performs similarly in many tasks with less complexity